

LEARNING WITH REGRESSION AND TREES

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Contents.....

- Simple Linear Regression
- Multiple Linear Regression
- Logistic Regression
- Regularization
- Evaluation Metric
- Probabilistic Model-Based Clustering
- BIRCH
- Fuzzy C-Means Use cases

Regression

“Regression analysis is a form of predictive modelling technique which investigates the relationship between a dependent and independent variable”

A regression analysis involves graphing a line over a set of data points that most closely fits the overall shape of the data. A regression shows the changes in a dependent variable on the y-axis to the explanatory variable on the x-axis.



Uses of Regression

Three major uses for regression analysis are

- Determining the strength of predictors
- Forecasting an effect, and
- Trend forecasting

3] Predicts trends and future values.
Regression analysis can be used to get point estimates. Questions like what will be the price of bitcoin in the next 6 months.

1] Regression might be used to identify the strength of the effect that the independent variables have on the dependent variables.

For e.g., what is the strength of the relationship between sales and marketing spending or what is the relationship between age and income.

2] Regression can be used to forecast effects or impact of changes.

In other words, we come to know how much the dependent variable changes with the change in one or more independent variables.

For e.g., how much additional sale income for each thousand dollars spent on marketing.

Types of Regression

- Based on the type of functions used to represent the relationship between the dependent or output variable and independent variables, the regression models are categorized into four types. The regression models are,
 1. Simple Linear Regression
 2. Multiple Regression
 3. Polynomial Regression
 4. Logistic Regression

SIMPLE LINEAR REGRESSION

Linear regression

- Linear regression is one of the easiest and most popular Machine Learning algorithms. It is a statistical method that is used for predictive analysis. Linear regression makes predictions for continuous/real or numeric variables such as **sales, salary, age, product price**, etc.
- Linear regression algorithm shows a linear relationship between a dependent (y) and one or more independent (x) variables, hence called as linear regression. Since linear regression shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.
- The linear regression model provides a sloped straight line representing the relationship between the variables.

Types of Regression

1. Simple Linear Regression

- Assume that there is only one independent variable x . If the relationship between x (independent variable) and y (dependent or output variable) is modeled by the relation,

$$y = a + bx$$

- then the regression model is called a linear regression model.

Types of Regression

2. Multiple Regression

- Assume that there are multiple independent variables *say* $x_1, x_2, \dots, x_n$. If the relationship between independent variables x and dependent or output variable y is modeled by the relation,

$$y = a_0 + a_1 * x_1 + a_2 * x_2 + \dots + a_n * x_n$$

Types of Regression

3. Polynomial regression

- Assume that there is only one independent variable x . If the relationship between independent variables x and dependent or output variable y is modeled by the relation,

$$y = a_0 + a_1 * x + a_2 * x^2 + \dots + a_n * x^n$$

- for some positive integer $n > 1$, then we have a polynomial regression.

Types of Regression

4. Logistic Regression

- Logistic regression is used when the dependent variable is binary (0/1, True/False, Yes/No) in nature.

Linear Regression

- When working with linear regression, our main goal is to find the best fit line that means the error between predicted values and actual values should be minimized. The best fit line will have the least error.
- The different values for weights or the coefficient of lines (β_0 , β_1) gives a different line of regression, so we need to calculate the best values for β_0 and β_1 to find the best fit line

Simple Linear Regression

- For simple linear regression, the form of the model is-

$$Y = \beta_0 + \beta_1 X$$

Where,

Y is a dependent variable.

X is an independent variable.

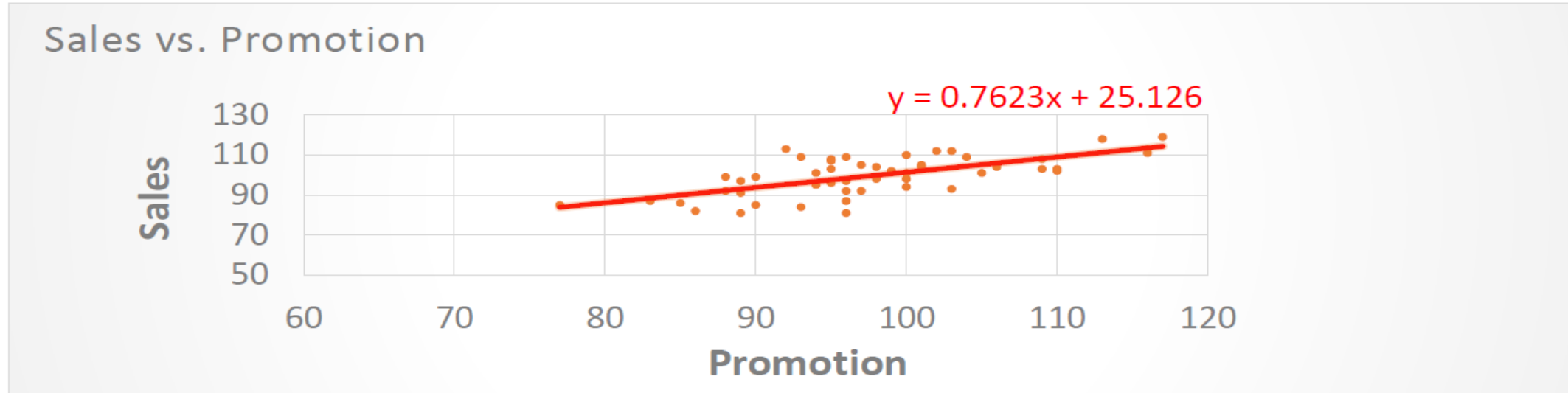
β_0 and **β_1** are the regression coefficients.

β_0 is the intercept or the bias that fixes the offset to a line.

β_1 is the slope or weight that specifies the factor by which X has an impact on Y.

Selecting Independent Variables: Scatter Plots

Scatter Plots are used to visualize the relationship between any two variables. For **regression analysis**, we are looking for **strong linear relationship** between the **independent** and **dependent** variable.



Selecting Independent Variables: Scatter Plots

- In statistical relationships, the plot is called a *scatter diagram* or *scatter plot* because of the scattering of points in the relation.
- The purpose of regression models is to identify a functional relationship between the target variable and a subset of the remaining attributes contained in the dataset.
- It enables to
 - Highlight and interpret the dependency of the target variable on other variables.
 - Predict the future value of the target variable

Simple Linear regression

**Simple
Linear
Regression**

$$y = b_0 + b_1 * x_1$$

Diagram illustrating the components of the Simple Linear Regression equation:

- Constant**: Points to b_0
- Coefficient**: Points to b_1
- Dependent variable (DV)**: Points to y
- Independent variable (IV)**: Points to x_1

Simple Linear Regression

$$b_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = 0$$

$$b_0 = \frac{1}{n} (\sum Y_i - b_1 \sum X_i) = \bar{Y} - b_1 \bar{X}$$

These are the **centered formulas** using means $\bar{X}, \bar{Y}$.

Simple Linear Regression

Previously Used Formula (Expanded Form):

$$\mathbf{a_1} \quad b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$\mathbf{a_0} \quad a = \frac{\sum y - b \sum x}{n}$$

- $y = a_0 + a_1 * x + e$

- *where*

- $a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{\overline{x^2} - \bar{x}^2}$

- $a_0 = \bar{y} - a_1 * \bar{x}$

These are the **expanded forms** derived algebraically from the centered formulas. They're easier to compute without calculating means directly.

Linear Regression

x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

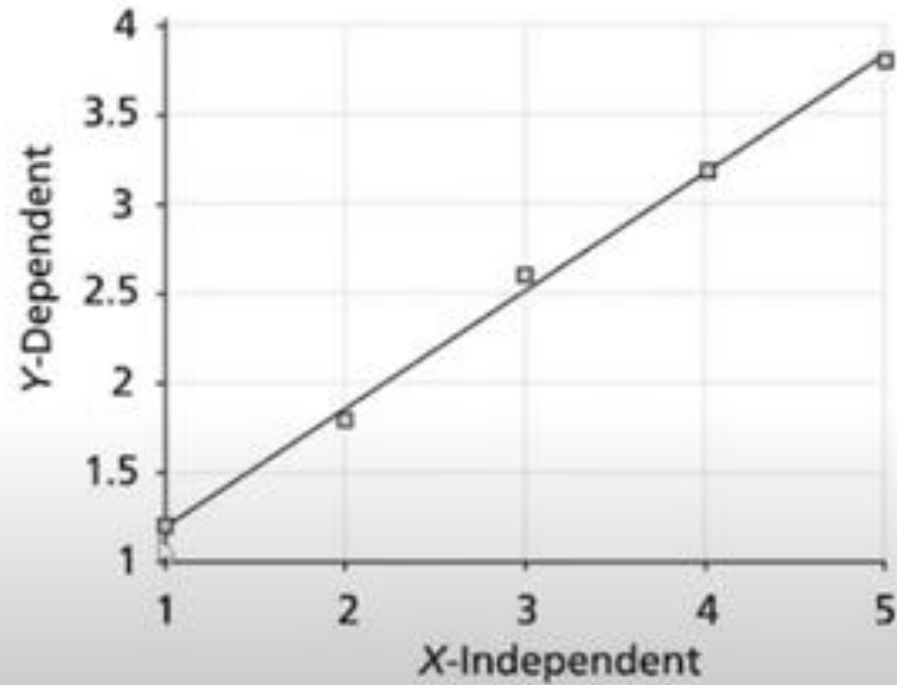
Solved
Example

Linear Regression

- Let us consider an example where the five weeks' sales data (in Thousands) is given as shown in Table.
- Apply linear regression technique to predict the 7th and 12th week sales.

x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Linear Regression



x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Linear Regression

X_i	Y_j	$X_i - \bar{X}$	$Y_j - \bar{Y}$	$(X_i - \bar{X}) * (Y_j - \bar{Y})$	$(X_i - \bar{X})^2$
1	1.2	-2	-1.32	2.64	4
2	1.8	-1	-0.72	0.72	1
3	2.6	0	0.08	0	0
4	3.2	1	0.68	0.68	1
5	3.8	2	1.28	2.56	4
X bar	Y bar			Sum	Sum
3	2.52			6.6	10

Linear Regression

$$b_1 = \frac{6.6}{10} = 0.66$$

Calculate b_0

$$b_0 = \bar{Y} - b_1 \cdot \bar{X} = 2.52 - (0.66)(3) = 2.52 - 1.98 = 0.54$$

Final Regression Equation:

$$Y = 0.54 + 0.66X$$

Linear Regression (using expanded formula)

- Linear regression equation is

given by

- $y = a_0 + a_1 * x + e$

- where

- $$a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{\overline{x^2} - \bar{x}^2}$$

- $$a_0 = \bar{y} - a_1 * \bar{x}$$

x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Linear Regression

- Linear regression equation is given by

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x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

- Here, there are 5 items, i.e., $i = 1, 2, 3, 4, 5$.

	x_i (Week)	y_j (Sales in Thousands)	x_i^2	$x_i * y_j$
	1	1.2	1	1.2
	2	1.8	4	3.6
	3	2.6	9	7.8
	4	3.2	16	12.8
	5	3.8	25	19

Linear Regression

- $\bar{x} = 3$ $\bar{y} = 2.52$ $\overline{x^2} = 11$ $\overline{xy} = 8.88$

- $a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{\overline{x^2} - \bar{x}^2} = \frac{8.88 - 3 \cdot 2.52}{11 - 3^2} = 0.66$

- $a_0 = \bar{y} - a_1 * \bar{x} = 2.52 - 0.66 * 3 = 0.54$

- Regression equation is

- $y = a_0 + a_1 * x$

- $y = 0.54 + 0.66 * x$

- Here, there are 5 items, i.e., $i = 1, 2, 3, 4, 5$.

	x_i (Week)	y_i (Sales in Thousands)	x_i^2	$x_i * y_i$
	1	1.2	1	1.2
	2	1.8	4	3.6
	3	2.6	9	7.8
	4	3.2	16	12.8
	5	3.8	25	19
Sum	15	12.6	55	44.4
Average	$\bar{x} = 3$	$\bar{y} = 2.52$	$\overline{x^2} = 11$	$\overline{xy} = 8.88$

Linear Regression

- $\bar{x} = 3$ $\bar{y} = 2.52$ $\overline{x^2} = 11$ $\overline{xy} = 8.88$

- $a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{\overline{x^2} - \bar{x}^2} = \frac{8.88 - 3 \cdot 2.52}{11 - 3^2} = 0.66$

- $a_0 = \bar{y} - a_1 * \bar{x} = 2.52 - 0.66 * 3 = 0.54$

- Regression equation is

- $y = a_0 + a_1 * x$

- $y = 0.54 + 0.66 * x$

- Regression equation is

- $y = a_0 + a_1 * x$

- $y = 0.54 + 0.66 * x$

- The predicted 7th week sale (when $x = 7$) is,

- $y = 0.54 + 0.66 * 7 = 5.16$

- the predicted 12th week sale (when $x = 12$) is,

- $y = 0.54 + 0.66 * 12 = 8.46$

Linear Regression

Linear Regression

x_i (Week)	y_j (Sales in Thousands)
1	1
2	3
3	4
4	8

Solved
Example
Matrix
Method

Linear Regression

- Find linear regression of the data of week and product sales (in Thousands) given in Table.
- Use linear regression in matrix form.
- Predict the 5th week sales.

x_i (Week)	y_j (Sales in Thousands)
1	1
2	3
3	4
4	8

Linear Regression

- Find linear regression of the data of week and product sales (in Thousands) given in Table.
- Use linear regression in matrix form.
- Predict the 5th week sales.

x_i (Week)	y_j (Sales in Thousands)
1	1
2	3
3	4
4	8

- Here, the independent variable X is be given as:

- $X^T = [1 \ 2 \ 3 \ 4]$

- The dependent variable is given as follows:

- $Y^T = [1 \ 3 \ 4 \ 8]$

- The data can be given in matrix form as

follows:

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 8 \end{pmatrix}$$

x_i (Week)	y_j (Sales in Thousands)
1	1
2	3
3	4
4	8

The first column can be used for setting bias.

Apply the Normal Equation

$$\theta = (X^T X)^{-1} X^T Y$$

► Compute X^T (transpose of X):

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

► Compute $X^T X$:

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

Apply the Normal Equation

$$\theta = (X^T X)^{-1} X^T Y$$

Compute $X^T Y$:

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 16 \\ 46 \end{bmatrix}$$

Solve for $\theta = (X^T X)^{-1} X^T Y$

We already have:

$$X^T X = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}, \quad X^T Y = \begin{bmatrix} 16 \\ 51 \end{bmatrix}$$

Solve for $\theta = (X^T X)^{-1} X^T Y$

We already have:

$$X^T X = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}, \quad X^T Y = \begin{bmatrix} 16 \\ 51 \end{bmatrix}$$

Find $(X^T X)^{-1}$

Let: $A = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$

Find determinant:

$$\det(A) = 4 \cdot 30 - 10 \cdot 10 = 120 - 100 = 20$$

Now compute inverse:

$$A^{-1} = \frac{1}{20} \begin{bmatrix} 30 & -10 \\ -10 & 4 \end{bmatrix}$$

Multiply with $X^T Y$

$$\theta = A^{-1} X^T Y = \frac{1}{20} \begin{bmatrix} 30 & -10 \\ -10 & 4 \end{bmatrix} \begin{bmatrix} 16 \\ 51 \end{bmatrix}$$

Now compute:

- First row: $30 \cdot 16 - 10 \cdot 51 = 480 - 510 = -30$
- Second row: $-10 \cdot 16 + 4 \cdot 51 = -160 + 204 = 44$

So:

$$\theta = \frac{1}{20} \begin{bmatrix} -30 \\ 44 \end{bmatrix} = \begin{bmatrix} -1.5 \\ 2.2 \end{bmatrix}$$

- Final Linear Regression Equation: _____

$$\boxed{y = -1.5 + 2.2x}$$

Example: 2

Estimated
(or predicted)
Y value for
observation i

Estimate of
the regression
intercept

Estimate of the
regression slope

Value of X for
observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

SUBJECT	AGE X	GLUCOSE LEVEL Y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81
7	55	?

Xi	Yj	Xi-Xbar	Yj-Ybar	(Xi-Xbar) * (Yj-Ybar)	(Xi-Xbar)^2
43	99	1.83	18	33.00	3.36
21	65	-20.17	-16	322.67	406.69
25	79	-16.17	-2	32.33	261.36
42	75	0.83	-6	-5.00	0.69
57	87	15.83	6	95.00	250.69
59	81	17.83	0	0.00	318.03
X bar	Y bar			Sum	Sum
41.2	81			478.00	1240.83

$$b1 = 478/1240 = 0.385$$

$$b0 = Ybar - b1 * Xbar = 65.138$$

$$y = b0 + b1x$$

$$\text{For } x=55, y = 65.138 + 0.385 * 55 = 86.313$$

Prediction – the value of y for the given value of x = 55

$$y' = 65.14 + (0.385225 * x)$$

$$y' = 65.14 + (.385225 * 55)$$

$$y' = 86.327$$

SUBJECT	AGE X	GLUCOSE LEVEL Y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81
7	55	86.327

Important points about LR

1. It is more susceptible to outliers hence;
2. It should not be used in the case of big-size data.
3. There should be a linear relationship between independent and dependent variables.
4. There is only one independent and dependent variable.
5. The type of regression line: a best fit straight line.

Advantages And Disadvantages of LR

Advantages	Disadvantages
Linear regression performs exceptionally well for linearly separable data	The assumption of linearity between dependent and independent variables

Advantages And Disadvantages of LR

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Linear regression performs exceptionally well for linearly separable data	The assumption of linearity between dependent and independent variables
Easier to implement, interpret and efficient to train	It is often quite prone to noise and overfitting

Advantages And Disadvantages of LR

Advantages	Disadvantages
Linear regression performs exceptionally well for linearly separable data	The assumption of linearity between dependent and independent variables
Easier to implement, interpret and efficient to train	It is often quite prone to noise and overfitting
It handles overfitting pretty well using dimensionally reduction techniques, regularization, and cross-validation	Linear regression is quite sensitive to outliers Hence,it should not be used in the case of big-size data

Use Case – Implementing Linear Regression

- 1.Loading the Data
- 2.Exploring the Data
- 3.Slicing The Data
- 4.Train and Split Data
- 5.Generate The Model
- 6.Evaluate The accuracy

MULTI-VARIANT LINEAR REGRESSION

Multiple linear regression

- is used to estimate the relationship between **two or more independent variables** and **one dependent variable**
- **Example:**
 - 1 The selling price of a house can depend on the desirability of the location, the number of bedrooms, the number of bathrooms, the year the house was built, the square footage of the lot and a number of other factors
 - 2 The height of a child can depend on the height of the mother, the height of the father, nutrition, and environmental factors.

Multiple Linear Regression

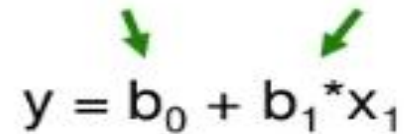
- Multiple Linear Regression attempts to model the relationship between two or more features and a response by fitting a linear equation to observed data.
- The steps to perform multiple linear Regression are almost similar to that of simple linear Regression.
- The Difference Lies in the evaluation.
- We can use it to find out which factor has the highest impact on the

$$\text{Here : } Y = b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_3 + \dots b_n * x_n$$

Y = Dependent variable and $x_1, x_2, x_3, \dots x_n$ = multiple independent variables

Multi variable Linear regression

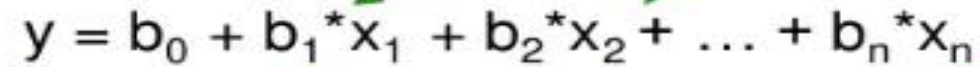
Simple Linear
Regression

$$y = b_0 + b_1 * x_1$$


Multiple Linear
Regression

Dependent variable (DV)

Independent variables (IVs)

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n$$


Constant

Coefficients

Multiple linear regression

Multiple Regression Model with k Independent Variables:

The diagram shows the equation $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$ with three labels in pink boxes above it. Arrows point from each label to its corresponding term in the equation: 'Y-intercept' points to β_0 , 'Population slopes' points to β_1 , β_2 , and β_k , and 'Random Error' points to ϵ .

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

Labels and their corresponding terms:

- Y-intercept: β_0
- Population slopes: β_1 , β_2 , β_k
- Random Error: ϵ

Multiple Linear Regression

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$a = b_0 = Y - b_1 X_1 - b_2 X_2$$

Multiple linear regression

The simplest multiple regression model for two predictor variables is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

Regression Equation:

$$\hat{y} = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$$

2. Formula for Coefficient b_1

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

This tells us how much y changes when x_1 increases by 1 unit, keeping x_2 constant.

3. Formula for Coefficient b_2

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Similarly, this tells us how much y changes when x_2 increases by 1 unit, keeping x_1 constant.

Multiple linear regression

The simplest multiple regression model for two predictor variables is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

Predict the value of Y given X_1 and X_2

SUBJECT	Y	X_1	X_2
1	-3.7	3	8
2	3.5	4	5
3	2.5	5	7
4	11.5	6	3
5	5.7	2	1
6	?	3	2

Regression formulas.....

$$\sum x_1^2 = \sum X_1^2 - \frac{(\sum X_1)^2}{n}$$

$$\sum x_2^2 = \sum X_2^2 - \frac{(\sum X_2)^2}{n}$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{N}$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{N}$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{N}$$

Multiple Linear Regression (Example)

	Y	X ₁	X ₂	X ₁ ²	X ₂ ²	X ₁ *Y	X ₂ *Y	X ₁ *X ₂
	-3.7	3	8					
	3.5	4	5					
	2.5	5	7					
	11.5	6	3					
	5.7	2	1					
SUM =								
MEAN =								

Multiple Linear Regression (Example)

	Y	X ₁	X ₂	X ₁ ²	X ₂ ²	X ₁ *Y	X ₂ *Y	X ₁ *X ₂
	-3.7	3	8	9	64	-11.1	-29.6	24
	3.5	4	5	16	25	14	17.5	20
	2.5	5	7	25	49	12.5	17.5	35
	11.5	6	3	36	9	69	34.5	18
	5.7	2	1	4	1	11.4	5.7	2
SUM =	19.5	20	24	90	148	95.8	45.6	99
MEAN =		4	4.8					

Linear Regression 2 independent variable

$$\sum x_1^2 = \sum X_1 X_1 - \frac{(\sum X_1)(\sum X_1)}{N} = 10$$

$$\sum x_2^2 = \sum X_2 X_2 - \frac{(\sum X_2)(\sum X_2)}{N} = 32.8$$

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{N} = 17.8$$

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{N} = -48$$

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{N} = 3$$

$$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$$

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{32.8 * 17.8 - 3 * (-48)}{10 * 32.8 - 3 * 3} = 2.28$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{10 * (-48) - 3 * 17.8}{10 * 32.8 - 3 * 3} = -1.67$$

$$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2 = \frac{19.5}{5} - \frac{2.28 * 20}{5} - \frac{-1.67 * 24}{5} = 2.796$$

Predict the value of Y given X_1 and X_2

Final Regression equation or Model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

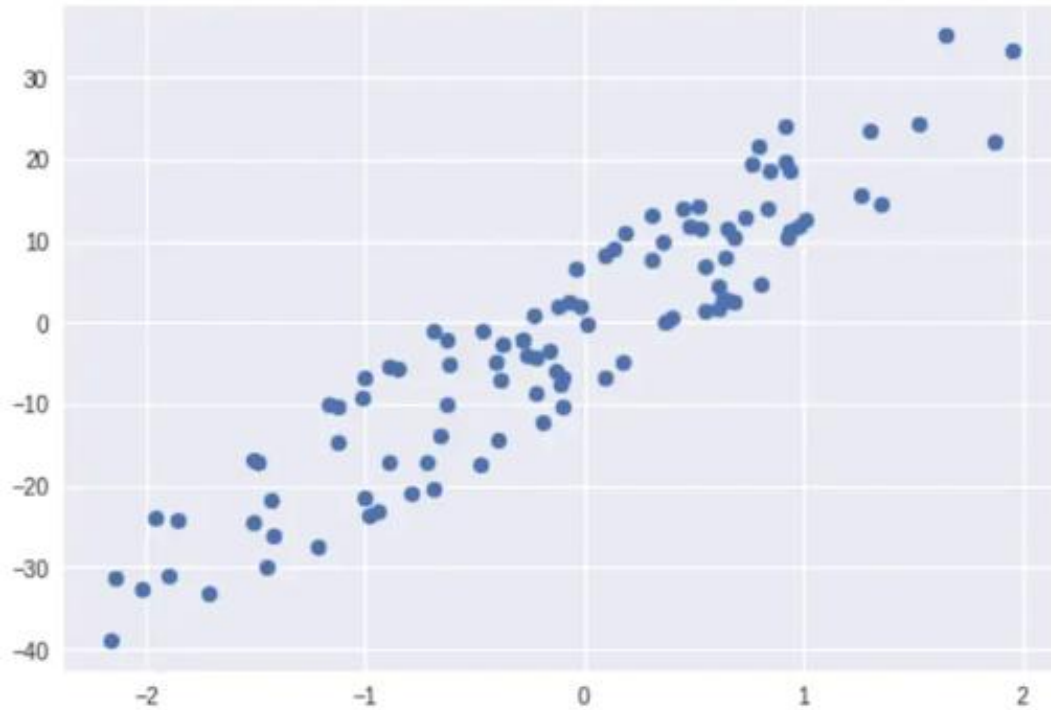
$$Y = 2.796 + 2.28 x_1 - 1.67 x_2$$

Now given $x_1 = 3$ and $x_2 = 2$ $Y = ?$

$$\begin{aligned} Y &= 2.796 + 2.28 * 3 - 1.67 * 2 \\ &= 6.296 \end{aligned}$$

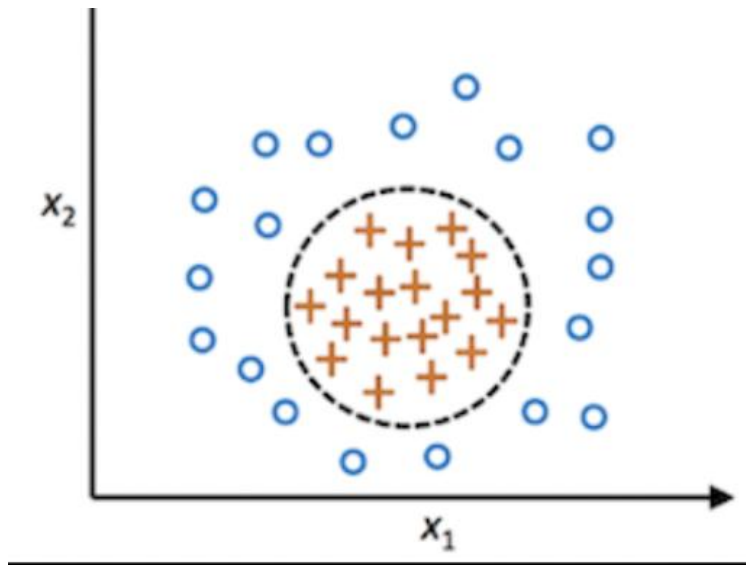
ECT	Y	X_1	X_2
	-3.7	3	8
	3.5	4	5
	2.5	5	7
	11.5	6	3
	5.7	2	1
	?	3	2

Linear data and its benefits



- Linear data is the type of data that follows a linear trend along the two-dimensional plane.
- The data points of linear data are connected sequentially.
- This sequential connection of data points helps traverse and interpret the result for each data point within the two-dimensional plane.

Non-linear data and its limitations



- The problem of non-linear regression can be solved by two methods:

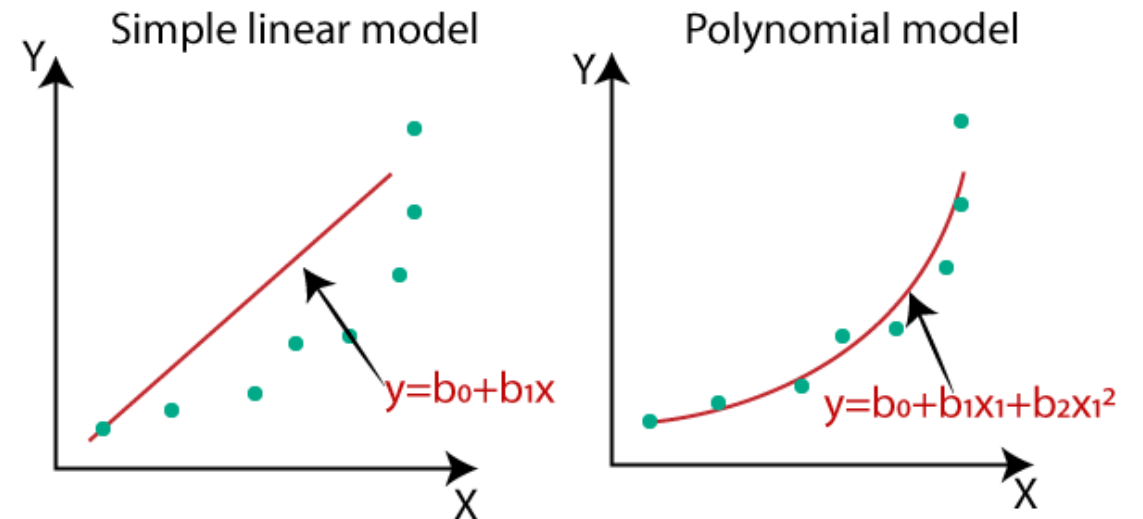
1. Transformation of non-linear data to linear data, so that the linear regression can handle the data
2. Using polynomial regression

Polynomial Regression

- It is also called the **special case of Multiple Linear Regression**.
- Because we add some polynomial terms to the Multiple Linear regression equation to convert it into Polynomial Regression.
- It is a **linear model** with some modification in order to increase the accuracy.
- The dataset used in Polynomial regression for training is of non-linear nature.
- It makes use of a linear regression model to fit the complicated and non-linear functions and datasets.

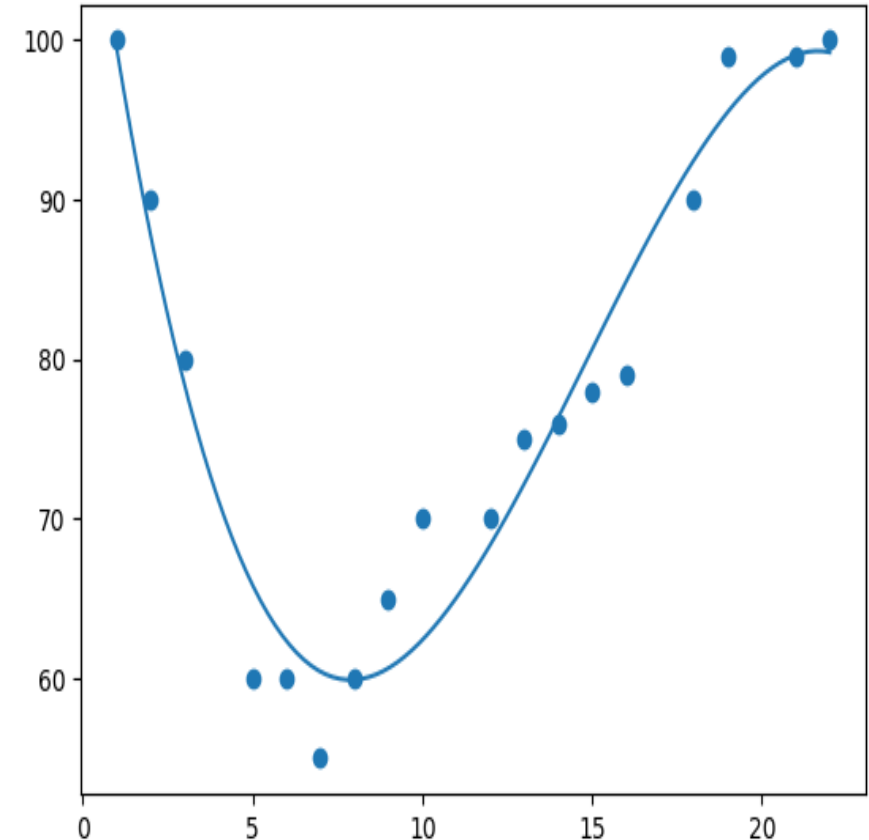
Need for Polynomial Regression

- If we apply a linear model on a **linear dataset**, then it provides us a good result as we have seen in Simple Linear Regression,
- but if we apply the same model without any modification on a **non-linear dataset**, then it will produce a drastic output.
- Due to which loss function will increase, the error rate will be high, and accuracy will be decreased.
- So for such cases, **where data points are arranged in a non-linear fashion, we need the Polynomial Regression model.**



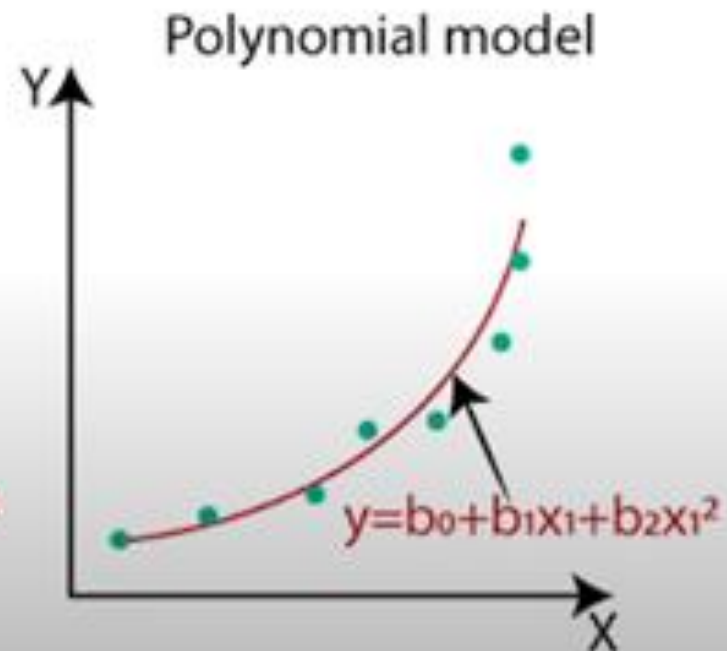
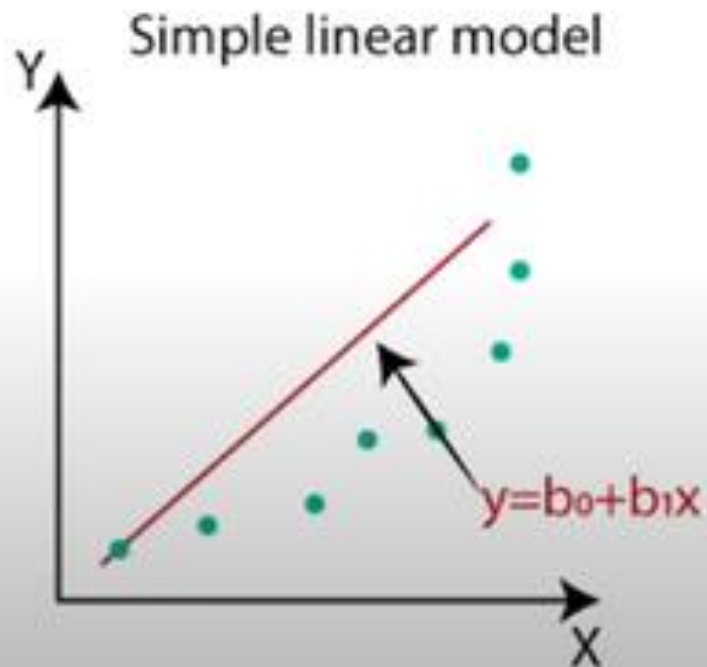
Equation of the Polynomial Regression Model

- Simple Linear Regression equation:
 - $y = b_0 + b_1x$
- Multiple Linear Regression equation:
 - $y = b_0 + b_1x + b_2x_2 + b_3x_3 + \dots + b_nx_n$
- Polynomial Regression equation:
 - $y = b_0 + b_1x + b_2x^2 + b_3x^3 + \dots + b_nx^n$
- The Simple and Multiple Linear equations are also Polynomial equations with a single degree, and the Polynomial regression equation is Linear equation with the nth degree.
- If your data points clearly will not fit a linear regression (a straight line through all data points), it might be ideal for polynomial regression.



Polynomial Regression

- If the relationship between the independent and dependent variables is not linear, then linear regression cannot be used as it will result in large errors.



Polynomial Regression

Polynomial Regression

- It can handle non-linear relationships among variables by using n^{th} degree of a polynomial.
- Instead of applying transformations, polynomial regression can be directly used to deal with different levels of curvilinearity.
- For example, the second-degree polynomial (called quadratic transformation) is given as:
 $y = a_0 + a_1x + a_2x^2$ and third degree polynomial is called cubic transformation given as: $y = a_0 + a_1x + a_2x^2 + a_3x^3$
- Generally, polynomials of maximum degree 4 are used, as higher order polynomials take some strange shapes and make the curve more flexible.
- It leads to a situation of overfitting and hence is avoided.

Polynomial Regression

- Consider the polynomial of 2nd degree.
- The polynomial equation is given by $y = a_0 + a_1x + a_2x^2$
- The coefficients a_0, a_1 and a_2 are calculated using the formula,

$$a = X^{-1}B$$

- Where,

$$X = \begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \quad B = \begin{bmatrix} \sum y_i \\ \sum (x_i, y_i) \\ \sum (x_i^2, y_i) \end{bmatrix}$$

Polynomial Regression

x	y
1	1
2	4
3	9
4	15

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- The polynomial equation is given by $y = a_0 + a_1x + a_2x^2$
- The coefficients a_0, a_1 and a_2 are calculated using the formula,

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x_i	y_i	$x_i y_i$	x_i^2	$x_i^2 y_i$	x_i^3	x_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
$\sum x_i = 10$	$\sum y_i = 29$	$\sum x_i y_i = 96$	$\sum x_i^2 = 30$	$\sum x_i^2 y_i = 338$	$\sum x_i^3 = 100$	$\sum x_i^4 = 354$

Polynomial Regression

X_i	Y_i	$X_i * Y_i$	X_i^2	$X_i^2 * Y_i$	X_i^3	X_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
Sum = X_i	Sum = Y_i	Sum = $X_i * Y_i$	Sum = X_i^2	Sum = $X_i^2 * Y_i$	Sum = X_i^3	Sum = X_i^4
10	29	96	30	338	100	354

Polynomial Regression

x	y
1	1
2	4
3	9
4	15

x_i	y_i	$x_i y_i$	x_i^2	$x_i^2 y_i$	x_i^3	x_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
$\Sigma x_i = 10$	$\Sigma y_i = 29$	$\Sigma x_i y_i = 96$	$\Sigma x_i^2 = 30$	$\Sigma x_i^2 y_i = 338$	$\Sigma x_i^3 = 100$	$\Sigma x_i^4 = 354$

$$a = X^{-1}B \quad X = \begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \quad B = \begin{bmatrix} \Sigma y_i \\ \Sigma(x_i, y_i) \\ \Sigma(x_i^2, y_i) \end{bmatrix} \quad \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{pmatrix} -0.75 \\ 0.95 \\ 0.75 \end{pmatrix}$$

Polynomial Regression

x_i	y_i	$x_i y_i$	x_i^2	$x_i^2 y_i$	x_i^3	x_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
$\Sigma x_i = 10$	$\Sigma y_i = 29$	$\Sigma x_i y_i = 96$	$\Sigma x_i^2 = 30$	$\Sigma x_i^2 y_i = 338$	$\Sigma x_i^3 = 100$	$\Sigma x_i^4 = 354$

$$\mathbf{a} = \mathbf{X}^{-1} \mathbf{B} \quad \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{pmatrix} -0.75 \\ 0.95 \\ 0.75 \end{pmatrix} \quad \underline{y = -0.75 + 0.95x + 0.75x^2}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$$

What are the Metrics used to Evaluate the performance of Regression Models in ML

▪ Linear Regression Metrics

- Consider the following training set Table for predicting the sales of the items.
- Consider two fresh items I6 and I7, whose actual values are 80 and 75, respectively.
- A regression model predicts the values of the items I6 and I7 as 75 and 85, respectively.
- Find MAE, MSE, RMSE, ReIMSE and CV.

x_i <i>Items</i>	y_j <i>(Sales)</i>
I1	80
I2	90
I3	100
I4	110
I5	120

Test Items	Actual Value	Predicted Value
I6	80	75
I7	75	85

What are the Metrics used to Evaluate the performance of Regression Models in ML

▪ Linear Regression Metrics

• Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=0}^{n-1} |y_i - \hat{y}_i|$$

$$\text{MAE} = \frac{1}{2} \times \left[|80 - 75| + |75 - 85| \right] = \frac{15}{2} = 7.5$$

x_i Items	y_j (Sales)
I1	80
I2	90
I3	100
I4	110
I5	120

Test Items	Actual Value	Predicted Value
I6	80	75
I7	75	85

What are the Metrics used to Evaluate the performance of Regression Models in ML

- Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2$$

$$\text{MSE} = \frac{1}{2} \times (|80 - 75|^2 + |75 - 85|^2) = \frac{125}{2} = 62.5$$

x_i Items	y_j (Sales)
I1	80
I2	90
I3	100
I4	110
I5	120

Test Items	Actual Value	Predicted Value
I6	80	75
I7	75	85

What are the Metrics used to Evaluate the performance of Regression Models in ML

- Root Mean Square Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{62.5} = 7.91$$

x_i Items	y_j (Sales)
I1	80
I2	90
I3	100
I4	110
I5	120

Test Items	Actual Value	Predicted Value
I6	80	75
I7	75	85

What are the Metrics used to Evaluate the performance of Regression Models in ML

- **Relative MSE**

$$\text{RelMSE} = \frac{\sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2}{\sum_{i=0}^{n-1} (y_i - \bar{y}_i)^2}$$

- For finding **RelMSE** and **CV**, the training table should be used to find the average of y .

x_i Items	y_i (Sales)
I1	80
I2	90
I3	100
I4	110
I5	120

Test Items	Actual Value	Predicted Value
I6	80	75
I7	75	85

What are the Metrics used to Evaluate the performance of Regression Models in ML

- **Relative MSE**

$$\text{RelMSE} = \frac{\sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2}{\sum_{i=0}^{n-1} (y_i - \bar{y}_i)^2}$$

- For finding **RelMSE** and **CV**, the training table should be used to find the average of y .

x_i Items	y_i (Sales)
I1	80
I2	90
I3	100
I4	110
I5	120

The average of y is $\frac{80 + 90 + 100 + 110 + 120}{5} = \frac{500}{5} = 100$.

$$\text{RelMSE} = \frac{(80 - 75)^2 + (75 - 85)^2}{(80 - 100)^2 + (75 - 100)^2} = \frac{125}{1025} = 0.1219$$

Test Items	Actual Value	Predicted Value
I6	80	75
I7	75	85

What are the Metrics used to Evaluate the performance of Regression Models in ML

- **Coefficient of Variation**

$$CV = \frac{RMSE}{\bar{y}}$$

$$CV = \frac{\sqrt{62.5}}{100} = 0.08$$

x_i Items	y_j (Sales)
I1	80
I2	90
I3	100
I4	110
I5	120

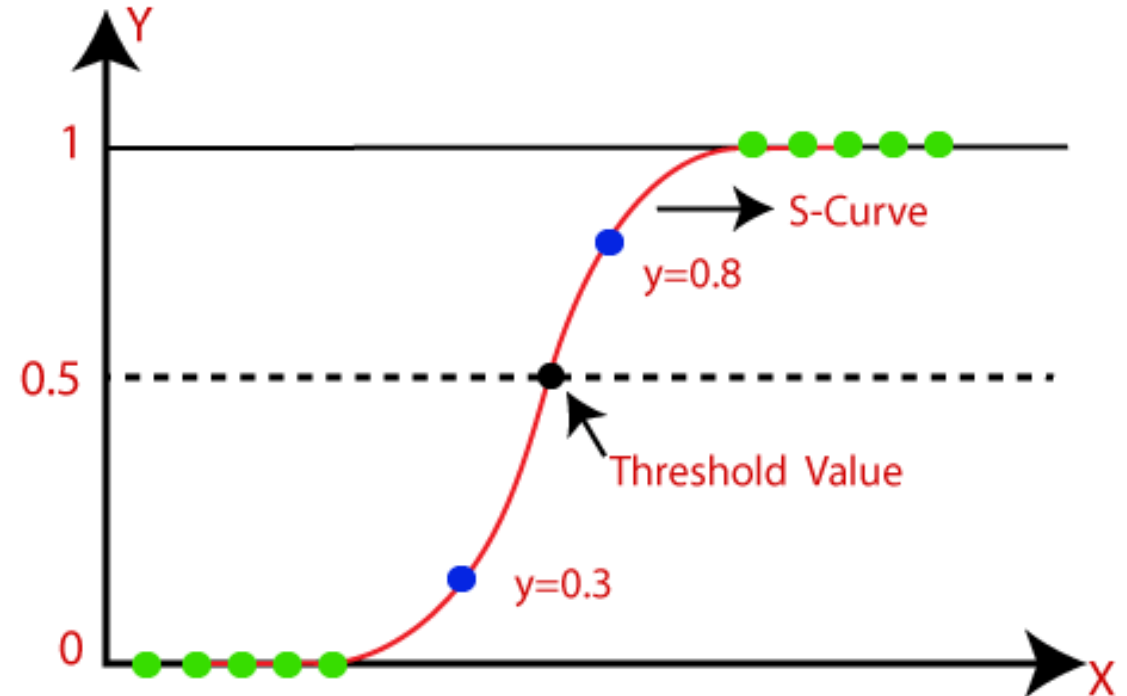
Test Items	Actual Value	Predicted Value
I6	80	75
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Logistic Regression

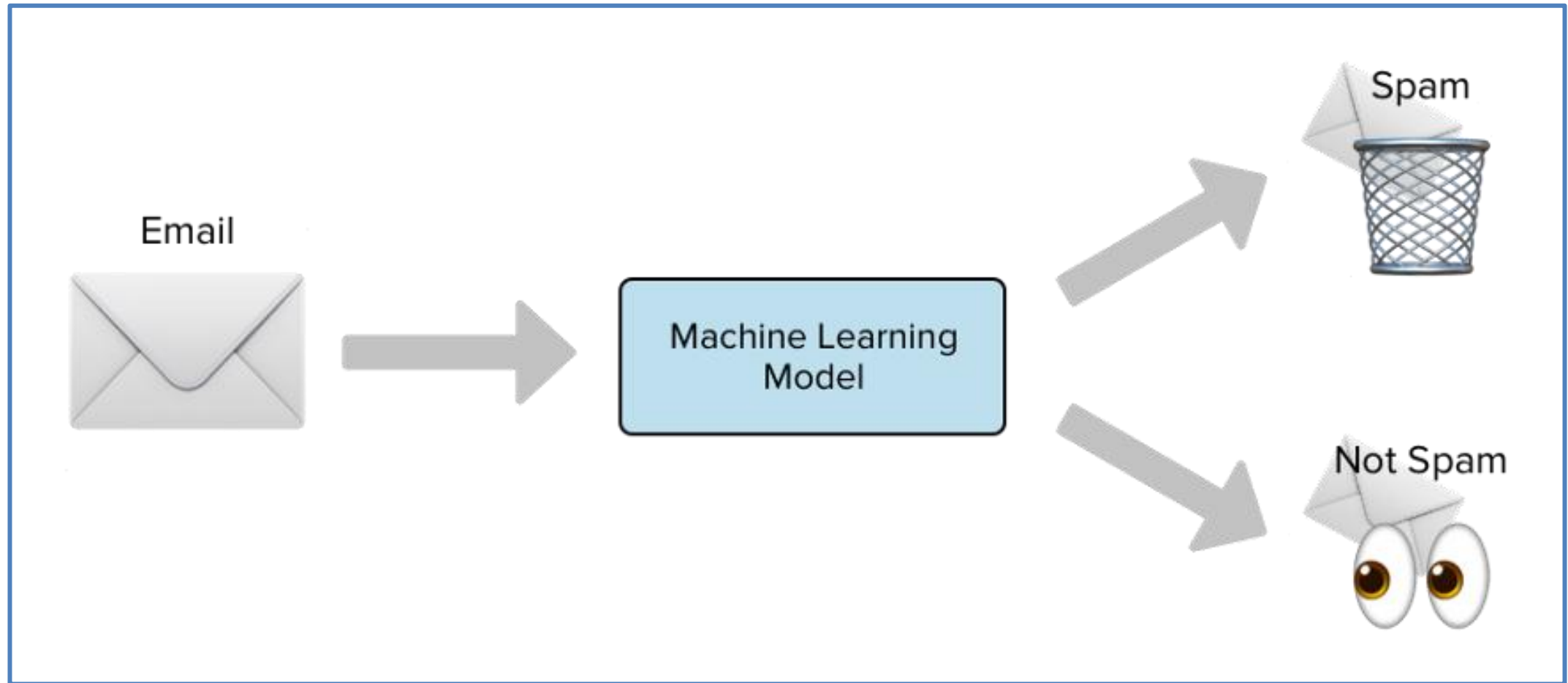
- Logistic regression is the Supervised Learning technique.
- Logistic regression is a classification algorithm.
- It is used for predicting the categorical dependent variable using a given set of independent variables.
- It is intended for datasets that have numerical input variables and a categorical target variable that has two values or classes.
- Logistic regression predicts the output of a categorical dependent variable.
- Therefore, the outcome must be a categorical or discrete value.
- It can be either Yes or No, 0 or 1, true or False, etc.
- but instead of giving the exact value as 0 and 1, **it gives the probabilistic values which lie between 0 and 1.**

Logistic Regression

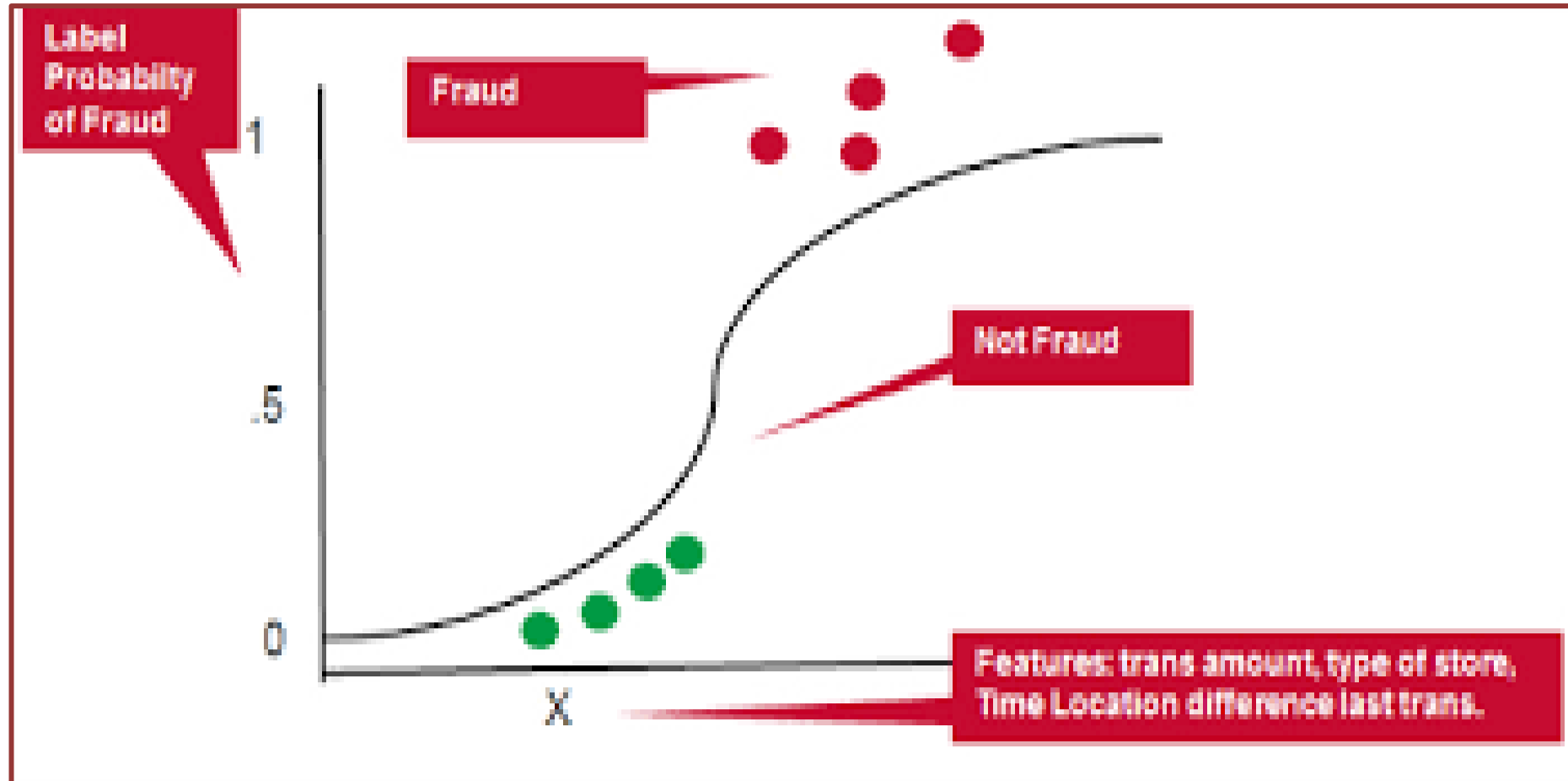
- Logistic Regression is much similar to the Linear Regression except that how they are used.
- Linear Regression is used for solving Regression problems, whereas **Logistic regression** is used for solving the **classification problems**.
- In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).
- The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not based on its weight, etc.



Applications of Logistic Regression

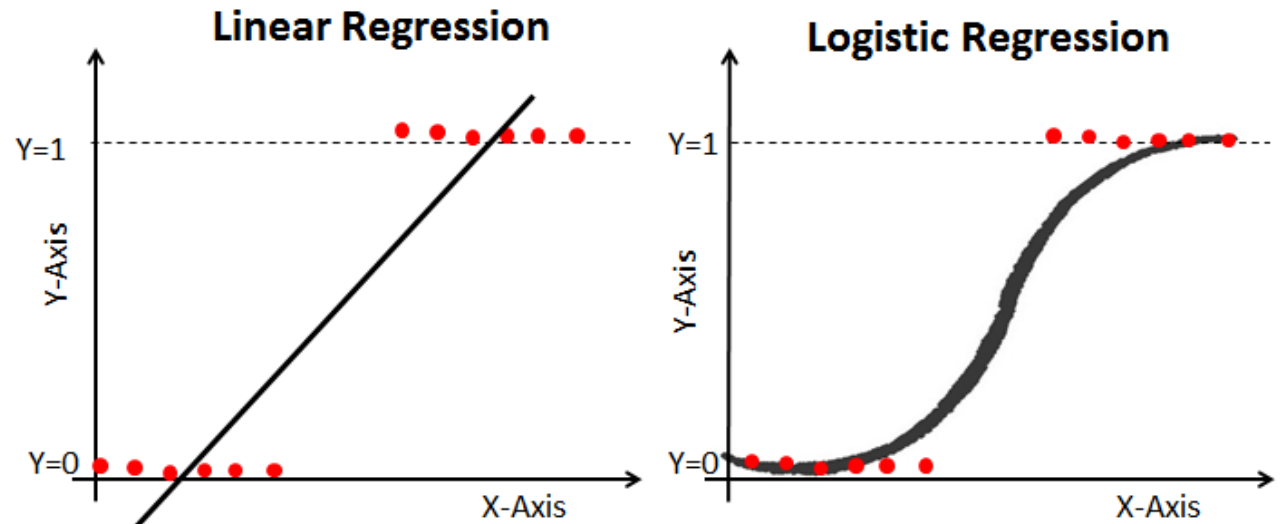


Applications of Logistic Regression



Linear vs Logistic Regression

- Linear regression gives you a continuous output, but logistic regression provides a constant output.
- An example of the continuous output is house price and stock price.
- Example of the discrete output is predicting whether a patient has cancer or not
- Linear regression is estimated using Ordinary Least Squares (OLS)
- while logistic regression is estimated using Maximum Likelihood Estimation (MLE) approach.



Logistic Regression Equation

- The mathematical steps to get Logistic Regression equations are given below:
- We know the equation of the straight line can be written as:

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$$

- In Logistic Regression y can be between 0 and 1 only, so for this let's divide the above equation by $(1-y)$:

$$\frac{y}{1-y} ; 0 \text{ for } y=0, \text{ and infinity for } y=1$$

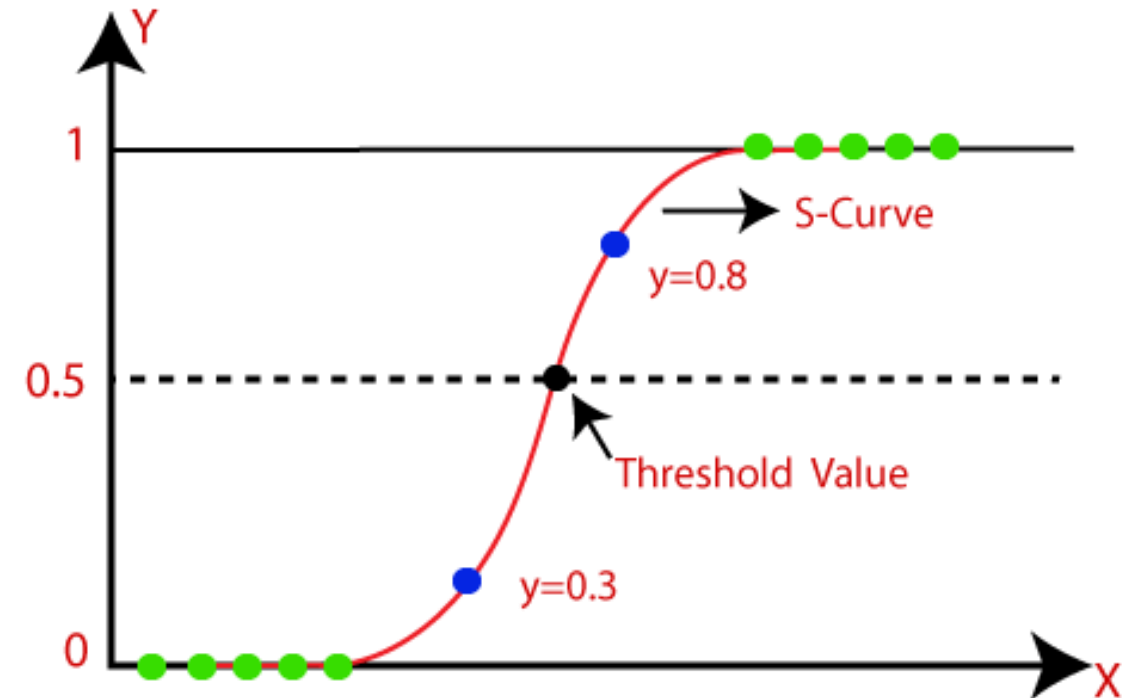
- But we need range between $-\infty$ to $+\infty$, then take logarithm of the equation it will become:

$$\log \left[\frac{y}{1-y} \right] = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$$

- The Logistic regression equation can be obtained from the Linear Regression equation.
 - $Y = 1 / (1 + e^{-x})$
 - Also called as **sigmoid function**

Logistic Function (Sigmoid Function)

- The sigmoid function is a mathematical function used to map the **predicted values to probabilities**.
- It maps any **real value into another value** within a range of 0 and 1.
- The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form. The S-form curve is called the Sigmoid function or the logistic function.
- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.



Type of Logistic Regression

Binomial

- only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.

Multinomial

- 3 or more possible unordered types of the dependent variable, such as "cat", "dogs", or "sheep"

Ordinal

- 3 or more possible ordered types of dependent variables, such as "low", "Medium", or "High".

Logistic Regression (Binomial)

- **Example:** There is a dataset given which contains the information of various users obtained from the social networking sites. There is a car making company that has recently launched a new SUV car. So the company wanted to check how many users from the dataset, wants to purchase the car.
- For this problem, we will build a Machine Learning model using the Logistic regression algorithm. The dataset is shown in the below image. In this problem, we will predict the **purchased variable (Dependent Variable)** by using **age and salary (Independent variables)**.

User ID	Gender	Age	EstimatedSalary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0
15728773	Male	27	58000	0
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	0
15727311	Female	35	65000	0
15570769	Female	26	80000	0
15606274	Female	26	52000	0
15746139	Male	20	86000	0
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1
15617482	Male	45	26000	1
15704583	Male	46	28000	1
15621083	Female	48	29000	1
15649487	Male	45	22000	1
15736760	Female	47	49000	1

Logistic Regression – Solved Example 1

- The dataset of pass or fail in an exam of 5 students is given in the table.
 - Use logistic regression as classifier to answer the following questions.
1. Calculate the probability of pass for the student who studied 33 hours.
 2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

Logistic Regression – Solved Example 1

- The dataset of pass or fail in an exam of 5 students is given in the table.
 - Use logistic regression as classifier to answer the following questions.
1. Calculate the probability of pass for the student who studied 33 hours.
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Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

Assume the model suggested by the optimizer for odds of passing the course is,

$$\log(odds) = -64 + 2 * hours$$

Logistic Regression – Solved Example 1

1. Calculate the probability of pass for the student who studied 33 hours.

$$\bullet \quad p = \frac{1}{1+e^{-z}} \quad s(x) = \frac{1}{1+e^{-x}}$$

$$\bullet \quad z = -64 + 2 * 33 = -64 + 66 = \textcircled{2}$$

$$\bullet \quad p = \frac{1}{1+e^{-2}} = \underline{0.88}$$

• That is, if student studies 33 hours, then there is **88% chance** that the student will pass the exam

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

$$\log(\text{odds}) = z = \underline{-64 + 2 * \text{hours}}$$

Logistic Regression – Solved Example 1

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

- $p = \frac{1}{1+e^{-z}} = 0.95$
- $0.95 * (1 + e^{-z}) = 1$
- $0.95 * e^{-z} = 1 - 0.95$
- $e^{-z} = \frac{0.05}{0.95} = 0.0526$ ✓
- $\ln(e^{-z}) = \ln(0.0526)$

$$\ln(e^x) = x$$

$$-z = \ln(0.0526) = -2.94$$

$$z = 2.94$$

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

Logistic Regression – Solved Example 1

- $z = 2.94$
- $\log(\text{odds}) = z = -64 + 2 * \text{hours}$
- $2.94 = -64 + 2 * \text{hours}$
- $2 * \text{hours} = \underline{2.94 + 64}$
- $2 * \text{hours} = \underline{66.94}$
- $\text{hours} = \frac{66.94}{2}$
- **$\text{hours} = 33.47 \text{ Hours}$**

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- The student should study **at least 33.47 hours**, so that he will pass the exam with more than 95% probability

Difference between Linear Regression and Logistic Regression

Sr No	Linear Regression	Logistic Regression
1	Linear Regression is used for solving Regression problem.	Logistic regression is used for solving Classification problems.
2	In Linear regression, we predict the value of continuous variables.	In logistic Regression, we predict the values of categorical variables.
3	In linear regression, we find the best fit line, by which we can easily predict the output.	In Logistic Regression, we find the S-curve by which we can classify the samples.
4	Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
5	The output for Linear Regression must be a continuous value, such as price, age, etc.	The output of Logistic Regression must be a Categorical value such as 0 or 1, Yes or No, etc.
6	In Linear regression, it is required that relationship between dependent variable and independent variable must be linear.	In Logistic regression, it is not required to have the linear relationship between the dependent and independent variable.

Difference between Linear Regression and Logistic Regression

Sr. No	Linear Regression	Logistic Regression
7	Here no activation function is used.	Here activation function is used to convert a linear regression equation to the logistic regression equation
8	Here no threshold value is needed.	Here a threshold value is added.
9	Here we calculate Root Mean Square Error(RMSE) to predict the next weight value.	Here we use precision to predict the next weight value.
10	Applications of linear regression: • Financial risk assessment • Business insights • Market analysis	Applications of logistic regression: • Medicine • Credit scoring • Hotel Booking • Gaming • Text editing

Advantages of Logistic Regression

1. Logistic Regression is very easy to understand.
2. It requires less training.
3. It performs well for simple datasets as well as when the data set is linearly separable.
4. It doesn't make any assumptions about the distributions of classes in feature space.
5. A Logistic Regression model is less likely to be over-fitted but it can overfit in high dimensional datasets. To avoid over-fitting these scenarios, One may consider regularization.
6. They are easier to implement, interpret, and very efficient to train.

Disadvantages of Logistic Regression

1. Sometimes a lot of **Feature Engineering** is required.
2. If the independent features are correlated with each other, it may affect the performance of the classifier.
3. It is quite sensitive to **noise** and **overfitting**.
4. Logistic Regression should not be used if the number of observations is lesser than the number of features, otherwise, it may lead to overfitting.
5. By using Logistic Regression, non-linear problems can't be solved because it has a linear decision surface. But in real-world scenarios, the linearly separable data is rarely found.
6. By using Logistic Regression, it is tough to obtain complex relationships. Some algorithms such as neural networks, which are more powerful, and compact can easily outperform Logistic Regression algorithms.
7. In Linear Regression, there is a linear relationship between independent and dependent variables but in Logistic Regression, independent variables are linearly related to the log odds ($\log(p/(1-p))$).

Questions on Logistic Regression?

- **What are the different types of Logistic Regression?**
 1. **Binary Logistic Regression:** In this, the target variable has only two 2 possible outcomes.
 - **For Example,** 0 and 1, or pass and fail or true and false.
 2. **Multinomial Logistic Regression:** In this, the target variable can have three or more possible values without any order.
 - **For Example,** Predicting preference of food i.e. Veg, Non-Veg, Vegan.
 3. **Ordinal Logistic Regression:** In this, the target variable can have three or more values with ordering.
 - **For Example,** Movie rating from 1 to 5.

Linear Regression Use Cases

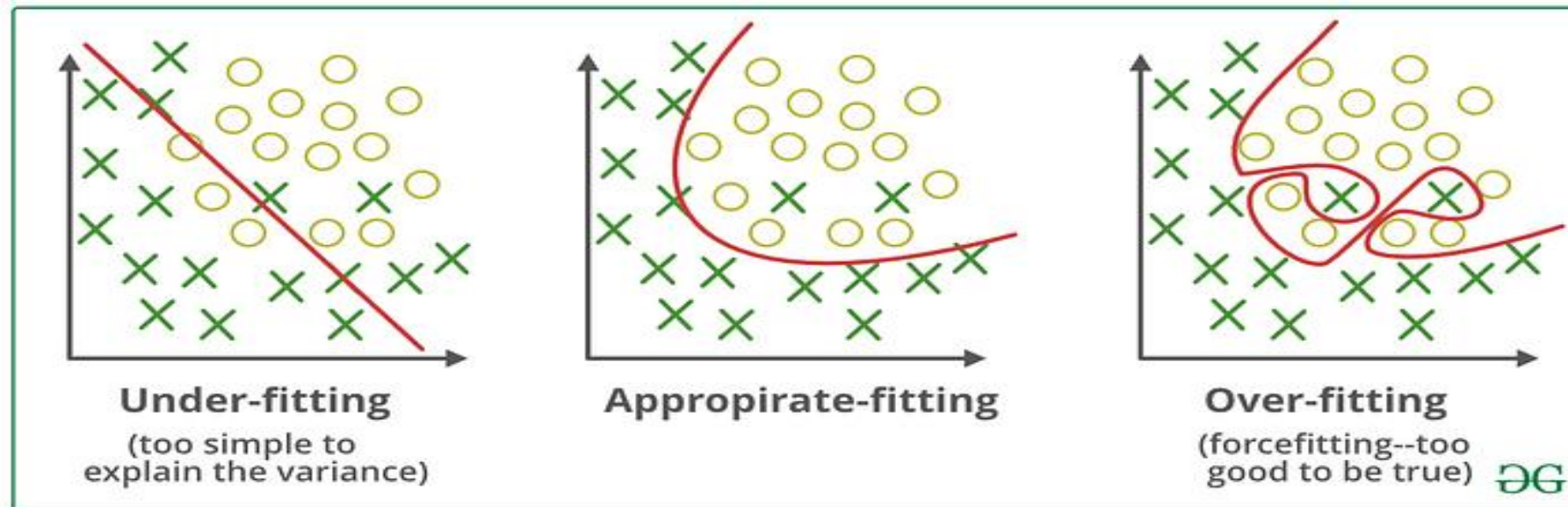
- Sales of a product; pricing, performance, and risk parameters
- Generating insights on consumer behavior, profitability, and other business factors
- Evaluation of trends; making estimates, and forecasts
- Determining marketing effectiveness, pricing, and promotions on sales of a product
- Assessment of risk in financial services and insurance domain
- Studying engine performance from test data in automobiles
- Calculating causal relationships between parameters in biological systems
- Conducting market research studies and customer survey results analysis
- Astronomical data analysis
- Predicting house prices with the increase in sizes of houses

Logistic Regression Use Cases

- Spam Detection
- Credit Card Fraud
- Marketing
- Medicine
- Text editing
- Hotel Booking
- Gaming

What is Regularization?

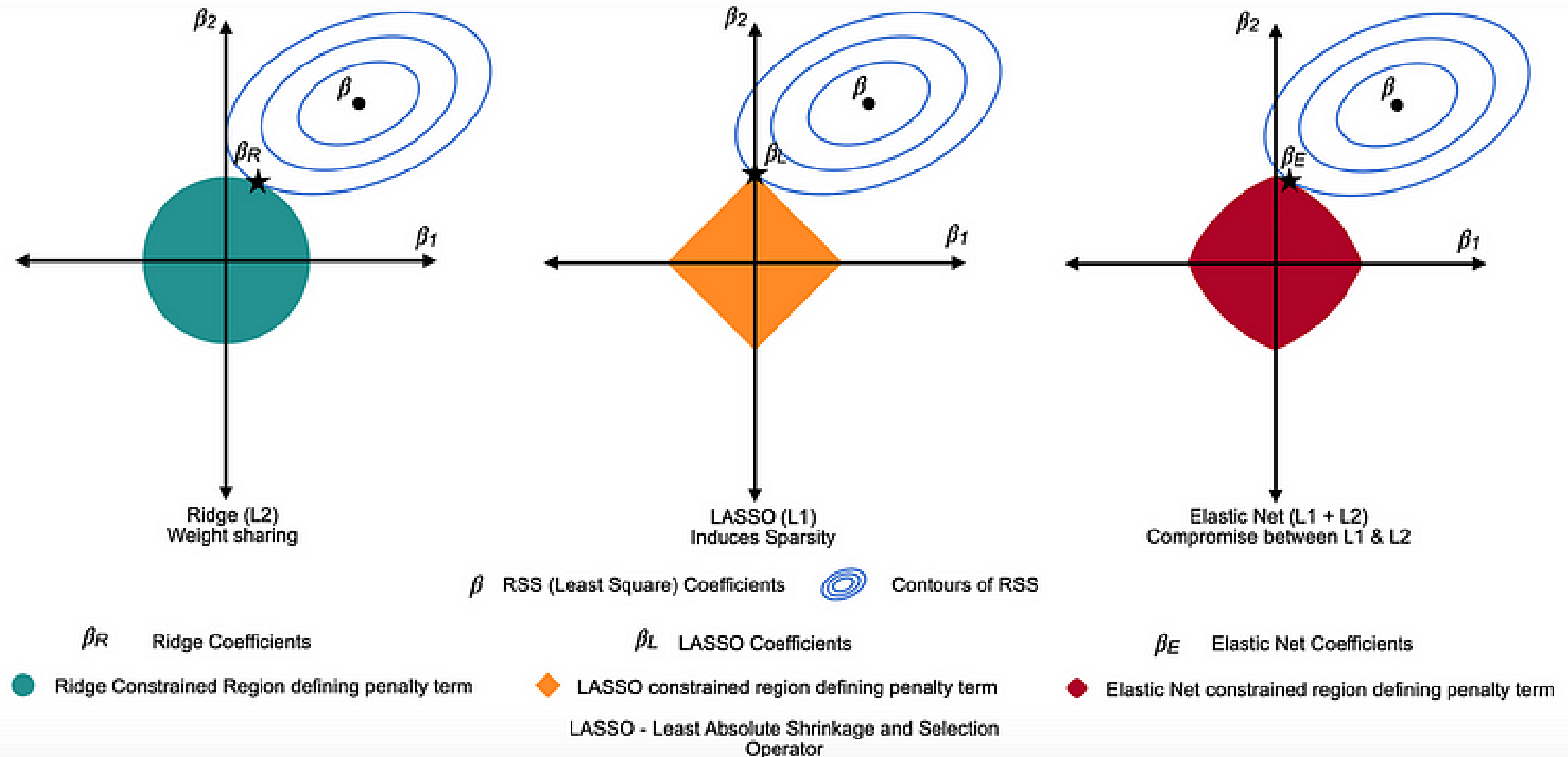
- Technique to prevent overfitting in machine learning models.
- Adds a penalty term to the objective function.
- Encourages simpler models that generalize better to unseen data.
- Balances model complexity and performance.



How Does Regularization Work?

- Regularization modifies the **loss function** by adding a **penalty term**.
- New Loss = Original Loss (e.g., MSE) + Regularization Term
- It **shrinks the coefficient values**, reducing model complexity.
- The strength of regularization is controlled by a parameter λ (lambda).
- Helps the model to avoid learning noise or irrelevant patterns.

Regularization Techniques:



Regularization in Machine Learning

- Regularization is a technique used to reduce errors by fitting the function appropriately on the given training set and avoiding overfitting. The commonly used [regularization techniques](#) are :
 - Lasso Regularization – L1 Regularization
 - Ridge Regularization – L2 Regularization
 - Elastic Net Regularization – L1 and L2 Regularization

Lasso Regression

- A regression model which uses the **L1 Regularization** technique is called **LASSO(Least Absolute Shrinkage and Selection Operator)** regression.
- **Lasso Regression** adds the “*absolute value of magnitude*” of the coefficient as a penalty term to the loss function(L).
- Lasso regression also helps us achieve feature selection by penalizing the weights to approximately equal to zero if that feature does not serve any purpose in the model.

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^m |w_i|$$

where,

- **m** – Number of Features
- **n** – Number of Examples
- **Y-i** – Actual Target Value
- **Y-i(hat)** – Predicted Target Value

Ridge Regression

- A regression model that uses the **L2 regularization** technique is called **Ridge regression**.
- **Ridge regression** adds the “*squared magnitude*” of the coefficient as a penalty term to the loss function(L).

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^m w_i^2$$

Elastic Net Regression

- This model is a combination of L1 as well as L2 regularization.
- That implies that we add the absolute norm of the weights as well as the squared measure of the weights.
- With the help of an extra [hyperparameter](#) that controls the ratio of the L1 and L2 regularization.

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \left((1 - \alpha) \sum_{i=1}^m |w_i| + \alpha \sum_{i=1}^m w_i^2 \right)$$

Key Differences between Ridge and Lasso Regression

Characteristic	Ridge Regression (L2 Regularization)	Lasso Regression (L1 Regularization):
Penalty Term:	Adds the sum of squared coefficients as a penalty term. $\text{Penalty Term} = \lambda \sum_{i=1}^n b_i^2$	Adds the sum of the absolute values of coefficients as a penalty term. $\text{Penalty Term} = \lambda \sum_{i=1}^n b_i $
Effect on Coefficients:	Shrinks the coefficients towards zero but does not set them exactly to zero. It is effective in preventing overfitting and handling multicollinearity.	Can shrink some coefficients exactly to zero, effectively performing feature selection. It not only prevents overfitting and handles multicollinearity but also provides a form of automatic feature selection.
Use Case:	1. Suitable when you believe that all features are important and should be included in the model. 2. Useful for handling multicollinearity (high correlation among independent variables).	1, Useful when you have a <u>large number of</u> features and believe that some of them are less important or irrelevant. 2. Effective for feature selection as it sets some coefficients to exactly zero.
Mathematical Expression:	Minimizes the cost function with the sum of squared coefficients. $\text{Cost} = \text{MSE} + \lambda \sum_{i=1}^n b_i^2$	Minimizes the cost function with the sum of absolute values of coefficients. $\text{Cost} = \text{MSE} + \lambda \sum_{i=1}^n b_i $